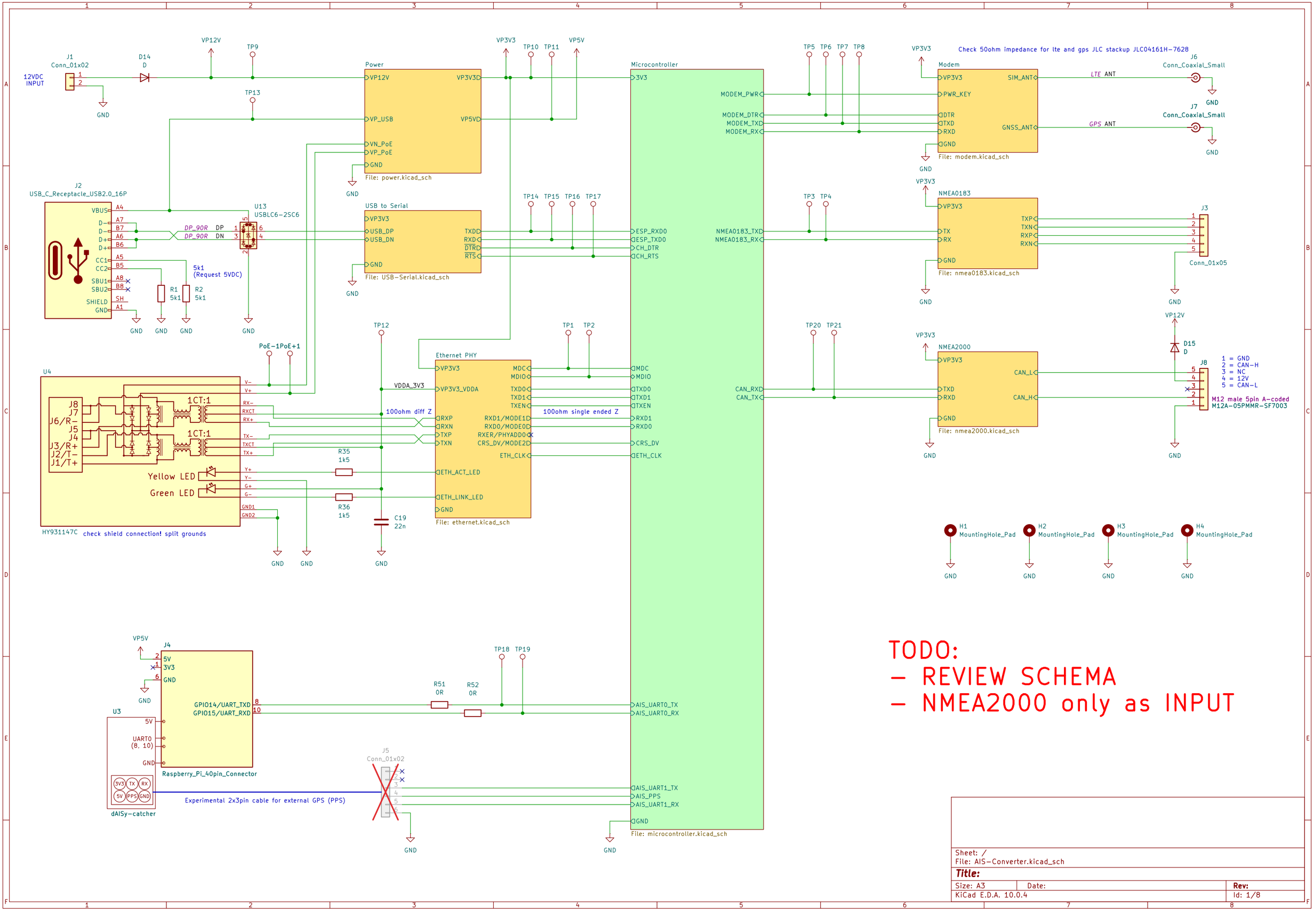
12VDC INPUT
J1
Conn_01x02
D14
D
VP12V
TP9
TP13
GND
Power
VP12V
VP3V3
VP_USB
VP5V
VN_PoE
VP_PoE
GND
File: power.kicad_sch
TP10
TP11
VP5V
VP3V3
Microcontroller
3V3
MODEM_PWR
MODEM_DTR
MODEM_TXD
MODEM_RXD
NMEA0183_TXD
NMEA0183_RXD
CAN_RXD
CAN_TXD
ESP_RXD0
ESP_TXD0
CH_DTR
CH_RTS
MDC
MDIO
TXD0
TXD1
TXEN
RXD1
RXD0
CRS_DV
ETH_CLK
ETH_ACT_LED
ETH_LINK_LED
GND
File: microcontroller.kicad_sch
TP5
TP6
TP7
TP8
Modem
VP3V3
PWR_KEY
DTR
TXD
RXD
GND
SIM_ANT
GNSS_ANT
File: modem.kicad_sch
Check 50ohm impedance for lte and gps JLC stackup JLC04161H-7628
LTE ANT
GPS ANT
J6
Conn_Coaxial_Small
J7
Conn_Coaxial_Small
GND
J2
USB_C_Receptacle_USB2.0_16P
VBUS
D-
D+
CC1
CC2
SBU1
SBU2
SHIELD
GND
U13
USBLC6-2SC6
DP_90R
DN_90R
DP
DN
5k1
(Request 5VDC)
R1
5k1
R2
5k1
USB to Serial
VP3V3
USB_DP
USB_DN
GND
TXD
RXD
DTR
RTS
File: USB-Serial.kicad_sch
TP14
TP15
TP16
TP17
TP3
TP4
NMEA0183
VP3V3
TX
RX
GND
TXP
TXN
RXP
RXN
File: nmea0183.kicad_sch
J3
Conn_01x05
TP20
TP21
NMEA2000
VP3V3
TXD
RXD
GND
CAN_L
CAN_H
File: nmea2000.kicad_sch
VP12V
D15
D
J8
1 = GND
2 = CAN-H
3 = NC
4 = 12V
5 = CAN-L
M12 male 5pin A-coded
M12A-05PMMR-SF7003
GND
TP12
TP1
TP2
Ethernet PHY
VP3V3
VP3V3_VDDA
MDC
MDIO
TXD0
TXD1
TXEN
RXP
RXN
TXP
TXN
RXD1/MODE1D
RXD0/MODE0D
RXER/PHYADD0
CRS_DV/MODE2D
ETH_CLK
ETH_ACT_LED
ETH_LINK_LED
GND
File: ethernet.kicad_sch
VDDA_3V3
100ohm diff Z
100ohm single ended Z
R35
1k5
R36
1k5
C19
22n
PoE-1PoE+1
U4
J8
J7
J6/R-
J5
J4
J3/R+
J2/T-
J1/T+
1CT:1
1CT:1
V-
V+
RX-
RXCT
RX+
TX-
TXCT
TX+
Y+
Y-
G+
G-
GND1
GND2
Yellow LED
Green LED
HY931147C
check shield connection! split grounds
H1
MountingHole_Pad
H2
MountingHole_Pad
H3
MountingHole_Pad
H4
MountingHole_Pad
VP5V
J4
5V
3V3
GND
GPIO14/UART_TXD
GPIO15/UART_RXD
Raspberry_Pi_40pin_Connector
U3
5V
UART0
(8, 10)
GND
3V3
TX
RX
5V
PPS
GND
dAISy-catcher
Experimental 2x3pin cable for external GPS (PPS)
R51
0R
R52
0R
TP18
TP19
J5
Conn_01x02
AIS_UART0_TX
AIS_UART0_RX
AIS_UART1_TX
AIS_PPS
AIS_UART1_RX
GND
TODO:
- REVIEW SCHEMA
- NMEA2000 only as INPUT
Sheet: /
File: AIS-Converter.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/8